Mittuniversitetet
MID SWEDEN UNIVERSITY

DT081A Signal and Image Processing

Exercise 1

1.1. Define mathematically a non-damped mechanical signal that oscillates around zero with amplitude of 20 cm, if it starts at 0 at time $t = 0$ and oscillates four periods.

1.2. Define mathematically a pure sinusoidal audio signal $s(t)$ that consists of 100 Hz, with zero phase-shift with an exponentially declining amplitude from 2 with a rate of $\sigma = 4$.

1.3. Formulate the mathematical description of the above audio signal being sampled at a sampling rate of $f_s = 1000$ Hz, i.e. $t = n \cdot T_s$, where
$T_s = 1/f_s$.

1.4. A digitized version of the signal in 1.1. can use 8 bits for the whole range to cover the signal amplitude values. Write the possible values of the signal amplitude.

1.5. A *digital* signal processing algorithm filters a signal $s(n), n = 0, \ldots$ with a filter g. Write the mathematical description and support Ω.

1.6. Describe the difference between audio processing and image processing with respect to dimensions, contents, and support.

1.7. Give examples of different Image Processing and where it could be applied.

1.8. What is the reason for using Red-Green-Blue representations of colour images?

1.9. Give possible applications of the examples given in the lecture for
1.9.1. Video processing?
1.9.2. Light field processing?
1.9.3. Plenoptic processing?

1.10. Describe the difference between Image Processing and Computer Vision.

1.11. What is the reason for introducing machine and deep learning algorithms instead of the classical image processing methods?

1.12. Based on the above answer, why should you also understand classical approaches for Image Processing?

1.13. Consider slide 33 of Lecture 1. What kind of processing do you envision at the different processing points, and what would the dimensionality of the data be?

1.14. Two complex numbers are $c_1 = -3 - j5$ and $c_2 = 5 + j3$.

1.14.1. Compute their amplitudes.

1.14.2. Compute their phases / arguments.

1.14.3. Express them in polar form.

1.14.4. Draw them and their complex conjugate in the complex plane. Indicate amplitude and phase for each of these vectors.

1.15. A time-discrete signal $s(n)$ consists of the following sequence: 2, 8, 3, -2, -1, 7 for $n = [0, 5]$. Write the signal in vector form.

1.16. The $s(n)$ above is filtered by g with the convolution $y(n) = \sum_{k=0}^{2} g_{n-k} s(k)$, where $g_k = \{1, -2, 2\}$ for $k = 0,1,2$
Write the convolution as a matrix multiplication for $y(n)$, $n = [0, 5]$.

1.17. A function is $x(t) = e^{-100t} u(t)$, where $u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$ is the heavy-side function (= step function).

1.17.1. Find the complex coefficients with the Hilbert space V: $\Phi = \{\varphi(t) = e^{j\omega t} \mid \omega \in \mathbb{R}, t \in \mathbb{R}\}$.

1.17.2. What is another name for this representation?

1.18. A time-discrete signal $s(n)$ consists of the following sequence: 1, -2, 2, 1 for $n = [0, 3]$.

1.18.1. Find the complex coefficients $X(e^{j\Omega})$ with the Hilbert space V: $\Phi = \{\varphi(n) = e^{j\Omega n} \,|\Omega \in \mathbb{R}, n \in \mathbb{Z}\}$.

1.18.2. What is another name for this representation?

1.19. Choose $N = 4$ values $\Omega_k,\ k = 0, \ldots N-1,$ equally spaced on the unit circle: $\Omega_k = \frac{2\pi k}{N}$.

1.19.1. Find the complex coefficients $X(k) = X(e^{j\Omega_k}),\ k = 0, \ldots N-1$ with the bases $\Phi = \{\varphi(n) = e^{j\Omega_k n} \,|k \in [0, N-1] \subset \mathbb{Z}, n \in \mathbb{Z}\}$ for the same signal as in 1.18.

1.19.2. What is another name for this representation?